

WQBEL BIOLOGIST RECOMMENDATIONS
FACT/DECISION SHEET
Cadillac WWTP - MI0020257 - REISSUANCE
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August 1, 2023

Background

The City of Cadillac submitted a National Pollutant Discharge Elimination System (NPDES) permit application on March 25, 2022, for reissuance of the Cadillac WWTP facility located at 1121 Plett Road, Cadillac, Michigan, 49601, Wexford County. The current NPDES permit was issued on August 30, 2019, effective on September 1, 2019, and expired on October 1, 2022, but has been extended.

The current NPDES permit authorizes to continuously discharge 3.2 million gallons per day (MGD) (this flow is not to be considered a limitation or actual capacity) of treated municipal wastewater from Monitoring Point 001A through Outfall 001. Outfall 001 discharges to the Clam River at Latitude 44.2657, Longitude -85.3985.

The Clam River has a lowest 95 percent exceedance, harmonic mean, and 90dQ10 low flows of 2.6, 13, and 3.4 cubic feet per second (cfs), respectively (Low Flow File No. 10194 [Wexford County]). The Clam River is designated as warmwater within this reach and is not protected as a public water supply source.

A total hardness value of 70 milligrams per liter (mg/l) was used for this review and was collected on August 19, 2021, upstream of Outfall 001 at Latitude 44.263376, Longitude -85.4011 (STORET Number 830054) by Permits Section staff.

Sources

Sources of information used for this review to develop the water quality-based effluent limits (WQBELs) include the facility's NPDES permit reissuance application, the current NPDES permit, the Permits Section Toxics facility file, final effluent monitoring report data for Perfluorooctane sulfonate (PFOS) from November 2017 through April 2023, and United States Geological Survey topographical maps. Based upon our review of this information, we have the following recommendations:

Recommendations for Outfall 001

- 1) At the request of staff from the Emerging Pollutants Section, we have reevaluated the Water Quality-Based Effluent Limitation (WQBEL) recommendations for Perfluorooctane sulfonate (PFOS). The current NPDES permit includes 2X annual monitoring for PFOS with samples collected as grab samples. Concentrations of PFOS in the Outfall 001 final effluent from November 2017 through April 2023 ranged from not detected (quantification level ranged from <1.6 ng/l to <1.9 ng/l) to 7.8 ng/l. The initial review for PFOS conducted in accordance with Rule 323.1211 indicates there is a reasonable potential for PFOS to be discharged from Outfall 001 at concentrations exceeding Michigan WQS.

The WQBEL review was then evaluated using the guidance in the May 29, 2020, Heaton memo titled "Water Resources Division (WRD) Guidance for Perfluorooctane sulfonate (PFOS) and Perfluorooctanoic Acid (PFOA) Reasonable Potential Evaluations." This guidance suggests taking the not detected results and entering the results as quantifiable values in the reasonable potential analysis.

Using the existing quantifiable results and applying the two not detected results as quantifiable values, there is not a reasonable potential for PFOS to be discharged from the Outfall 001 final effluent at concentrations exceeding Michigan Water Quality Standards (WQS). Based on this information, we have no recommendations for PFOS.

- a. The theoretical WQBEL for PFOS is a monthly average WQBEL of 0.012 ug/l (0.00032 lbs./day).
- 2) The recommendations for Perfluorooctanoic Acid (PFOA) and Perfluorobutanesulfonate (PFBS) in the review dated October 17, 2022, should remain as there is no information that would update these recommendations.

Other Recommendations

- 3) The recommendations contained herein are based on the Part 4 Water Quality Standards and Part 8 Water Quality-Based Effluent Limit Development for Toxic Substances. Treatment methods or economic concerns have not been addressed. We understand that other considerations may be used in deciding permit limits and requirements.